Remarks on Weak Solutions for Vortex Sheets with a Distinguished Sign

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Dedicated to Ciprian Foias with Admiration and Friendship

ABSTRACT. A simplified proof is presented here of J. M. Delort's recent existence theorem for two dimensional vortex sheets for initial vorticity of a single sign. The key simplification developed here involves an elementary but sharp a priori estimate for the decay of the vorticity maximal function valid only when the vorticity has a distinguished sign; this estimate is developed through use of physical conserved quantities for fluid flow involving the pseudo energy and second moments of vorticity. This new proof has enough flexibility to establish, as a corollary, that the high Reynolds number limit of Navier-Stokes solutions with vortex sheet initial data with a distinguished sign converges to a weak solution of the 2-D Euler equations.

1. Introduction. This paper concerns the existence of weak solutions of the two dimensional Euler equations with vortex sheet initial data. A two dimensional incompressible velocity field, $v_0 = (v_1^0, v_2^0)$, defines vortex sheet initial data provided that

$$\text{curl } v_0 = \omega_0 \text{ is a finite Radon measure, i.e., } \omega_0 \in M(R^2), \tag{1.1}$$

and

$$\begin{gathered} v_0 \text{ has locally finite kinetic energy, i.e.,} \\ \int_{|x|\le R} |v_0|^2\, dx \le C(R) < +\infty \text{ for any } R > 0. \end{gathered} \tag{1.2}$$

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Here, with the velocity, $v = (v_1, v_2)$, the scalar vorticity ω is defined by $\omega = \operatorname{curl} v = (v_2)_{x_1} - (v_1)_{x_2}$ and incompressibility is enforced by the condition, $\operatorname{div} v = 0$ where $\operatorname{div} v = (v_1)_{x_1} + (v_2)_{x_2}$. For simplicity in exposition, it is assumed throughout this paper that the initial vorticity, ω_0, has compact support, i.e.,

$$\operatorname{supp} \omega_0 \subseteq \{x \mid |x| < R_0\} \tag{1.3}$$

for some fixed $R_0 > 0$.

Vortex sheet initial data provide an excellent approximation for significant parts of the flow field in mixing layers, trailing wakes, etc., at high Reynolds numbers and there is an enormous engineering and applied mathematics literature devoted to building approximate solutions of the Euler equations with such initial data either through formal asymptotic methods ([1], [2]) or numerical computation ([3], [4]). In applications, the initial vorticity, ω_0, is typically a surface Dirac measure supported on a smooth curve. Classical linearized stability analysis of the simplest vortex sheets ([2]) reveals that the evolution of vortex sheets involves a classical Hadamard ill-posed initial value problem so that both computational and theoretical issues are extremely subtle. One large research effort ([2]) restricts the study of vortex sheets to analytic initial data and attempts to use complex analysis to study the first relatively faint onset of singularity formation. A completely different approach first emphasized by Diperna and the author ([5], [6], [7], [8]) circumvents the issues of ill-posedness and the stringent requirements of analyticity for short times by studying the existence of suitable weak (and measure-valued) solutions of the two dimensional Euler equations for all time. This point of view is motivated by the enormous complexity observed in careful ingenious numerical computations ([3], [4]) with vortex sheet initial data when the initial vorticity changes sign, i.e., the measure ω_0 has both positive and negative parts. New mathematical phenomena involving concentrations occur with such vortex sheet initial data ([6], [7], [8], [4]).

On the other hand, when the initial vorticity has a distinguished sign, i.e. without loss of generality,

$$\text{the measure } \omega_0 \text{ satisfies } \omega_0 \geq 0, \tag{1.4}$$

then despite the ill-posed character of these problems, both numerical computations ([3], [4]) and actual experiments ([9]) exhibit a complex but significantly more regular character in the solutions involving large amplitude Kelvin-Helmholtz rolls. Arguing formally, in two space dimensions the vorticity satisfies

$$\frac{D\omega}{Dt} = 0, \tag{1.5}$$

with $D/Dt = \partial/\partial t + (v_1)\partial/\partial x_1 + (v_2)\partial/\partial x_2$ so that $\omega_0 \geq 0$ guarantees $\omega \geq 0$ for all later times; since ω represents the infinitesmal rotation of the fluid elements, in the case of a distinguished sign for ω_0, all fluid elements spin in the same direction at all later times so that subtle cancellation and screening effects at small scales ([4]) are absent when the vorticity has a distinguished sign — the new features of the proofs presented in this paper translate this physical intuition into quantitative a priori estimates which confine the potential complexity in weak solutions when the vorticity has a distinguished sign. The striking difference in mathematical behavior for vortex sheets with a distinguished sign has been emphasized earlier by the author in [4] (p. 178) where Theorem 3.1 of [6] was utilized as an empirical test to check the strong convergence to weak solutions. More recently in [10], Delort has made an important contribution by proving that if the initial vortex sheet has a distinguished sign, $\omega_0 \geq 0$, then there is a weak solution for the 2-D Euler equations with the above initial data. The main goal of this paper is to present a simplified proof of Delort's theorem on vortex sheets. In the remainder of this introduction, I state in detail the results that are proved here and summarize the overall argument. The details are presented in the remainder of this paper.

As preliminary information (see [6]), I recall that an incompressible velocity field, $v(x,t) \in L^\infty\big([0,\infty), L^2_{\text{loc}}(R^2)\big)$ is a *weak solution* of the 2-D Euler equations with the initial data $v_0(x)$ provided that

(A) for all test functions $\Phi \in C_0^\infty(R^2 \times R^+)$ with $\operatorname{div}\Phi = 0$,

$$\iint (\Phi_t \cdot v + \nabla\Phi : v \otimes v)\, dx\, dt = 0; \tag{1.6}$$

(B) the velocity $v(x,t)$ belongs to $\operatorname{Lip}\big([0,\infty), H^{-L}_{\text{loc}}(R^2)\big)$ for some $L > 0$ and $v(0) = v_0$ in $H^{-L}_{\text{loc}}(R^2)$.

In (1.6), $v \otimes v = (v_i v_j)$, $\nabla\Phi = (\partial\Phi_i/\partial x_j)$, and $A : B$ denotes the matrix product $\sum_{i,j} a_{ij} b_{ij}$ while $H^s_{\text{loc}}(R^2)$ denotes the local L^2-Sobolev space with s-derivatives. The weak form in (1.6) arises in familiar fashion from writing the 2-D Euler equations in conservation form, multiplying by the test function Φ, and integrating by parts; the technical condition from (B) in (1.6) gives a precise meaning to the sense in which the initial data is achieved. In standard fashion ([6]), an incompressible velocity field can be recovered from a rapidly decreasing vorticity field via the two-dimensional Biot-Savart law through a stream function ψ; thus, the equations

$$\begin{aligned} \Delta\psi &= \omega, \\ v &= \nabla^\perp\psi = \begin{pmatrix} -\psi_{x_2} \\ \psi_{x_1} \end{pmatrix}, \end{aligned} \tag{1.7}$$

imply that

$$v(x) = \int_{R^2} K(x-y)\omega(y)\,dy \tag{1.8}$$

with

$$K(y) = \left(2\pi(y_1^2+y_2^2)\right)^{-1}\begin{pmatrix}-y_2\\ y_1\end{pmatrix} = \nabla^{\perp}\left(\frac{1}{2\pi}\log(|y|)\right).$$

Given vortex sheet initial data v_0, satisfying (1.1)–(1.3) with $\omega_0 \geq 0$, it is a simple matter to build smooth initial data, v_0^ε, approximating v_0 and satisfying the following requirements (see p. 316 of [6]).

(A) With $\omega_0^\varepsilon = \operatorname{curl} v_0^\varepsilon$, $\omega_0^\varepsilon \geq 0$ and $\operatorname{supp}\omega_0^\varepsilon \subseteq \{x \mid |x| < R_0\}$.
(B) For any $R > 0$,

$$\int_{|x|\leq R} |v_0^\varepsilon|^2\,dx + \int \omega_0^\varepsilon\,dx \leq C(R). \tag{1.9}$$

(C) ω_0^ε converges strongly to ω_0 in $H^{-1}_{\text{comp}}(R^2)$ and v_0^ε defined through ω_0^ε via (1.8) converges strongly to v_0 in $L^2_{\text{loc}}(R^2)$.

Consider the solution of the Navier-Stokes or Euler equations with the initial data v_0^ε, i.e., v^ε satisfies

$$\begin{aligned} &\frac{\partial v^\varepsilon}{\partial t} + v^\varepsilon\cdot\nabla v^\varepsilon = -\nabla p^\varepsilon + \nu_\varepsilon \Delta v^\varepsilon;\\ &\operatorname{div} v^\varepsilon = 0;\\ &v^\varepsilon|_{t=0} = v_0^\varepsilon. \end{aligned} \tag{1.10}$$

It is a well-known classical result ([11]) that the solutions v^ε exist and remain smooth for all positive times for any value of the viscosity ν_ε satisfying $\nu_\varepsilon \geq 0$. The main result which I discuss here is the following:

Theorem. *Consider any vortex sheet initial data v_0 with a distinguished sign for the vorticity, ω_0. Thus* (1.1)–(1.3) *are satisfied with $\omega_0 \geq 0$. Assume that the viscosity, ν_ε, is either identically zero* (*Euler limit*) *or that ν_ε converges to zero as $\varepsilon \to 0$ in some arbitrary fashion* (*Navier-Stokes limit*)*. Then, after passing to a subsequence, the smooth solutions, $v^\varepsilon(x,t)$, of the Navier-Stokes* (*Euler*) *equations in* (1.10) *converge weakly in $L^2_{\text{loc}}([0,\infty)\times R^2)$ to an incompressible velocity field, $v(x,t)$, which is a weak solution of the 2-D Euler equations satisfying* (1.6) *with the initial data $v_0(x)$.*

This theorem reduces to Delort's important result ([10]) in the case that the viscosity vanishes, i.e., $\nu_\varepsilon \equiv 0$. There is strong evidence ([12], [13]) through explicit examples for an analogous but simpler model problem involving the 1-D Vlasov-Poisson equations that the weak solutions in the theorem are probably not unique and depend on the regularization even though the vorticity has a distinguished sign. Next, I present a brief summary of the structure of the proof. The proof of the theorem has three main parts:

(1.11)

(1) Through the *special nonlinear structure of the 2-D Euler equations, it is sufficient to guarantee weak convergence for the special nonlinear functions* $v_1 v_2$, $v_1^2 - v_2^2$ in order to satisfy the requirement in (1.6) for a weak solution. (This fact was first developed and used by Diperna and the author on p. 71 of [7]).

(2) *A priori decay estimates for the vorticity maximal function when the vorticity has a distinguished sign.*

(3) *Delort's key observation* ([10]) *regarding the special nonlinearities* $v_1 v_2$ *and* $v_1^2 - v_2^2$ *and iterated mildly singular integrals.*

In Section 2 of this paper, I state the three ingredients from (1.11) in detail and combine them to yield the Theorem. Part (1) follows from a simple calculation which I repeat for the readers convenience while Part (3) is Delort's key observation regarding the special structure of these specific nonlinear terms. Part (2) in (1.11) is the simple new a priori estimate which simplifies Delort's proof in a straightforward fashion; the proof of this result is presented in Section 3. Finally, Section 4 contains some brief comments on open problems including the important case where the *vorticity measure does not have a distinguished sign.* It is worth mentioning here that Evans and Muller ([15]) have developed another "harmonic analysts" proof of Delort's theorem by utilizing the detailed properties of special quadratic functions operating on Hardy spaces; the proof in [15] is considerably more complex than Delort's original proof (and even more so for the argument presented here!) but does serve to elucidate some of the general features of the relevant nonlinear structure interacting with the requirement of nonnegative vorticity.

2. The three main steps and the proof of the Theorem. The first step in the proof from (1) in (1.11) involves exploiting the fact that the vector test functions Φ satisfy $\operatorname{div}\Phi = 0$. To account for this constraint, use a streamfunction, $\eta \in C_0^\infty\left(R^2 \times (0,\infty)\right)$ so that

$$\Phi = \nabla^\perp \eta = \begin{pmatrix} -\eta_{x_2} \\ \eta_{x_1} \end{pmatrix} \tag{2.1}$$

and insert this expression for Φ into (1.6); with straightforward algebraic simplification, one obtains that the formula in (1.6) reduces to

$$\begin{aligned}(2.2)\qquad &\iint \left[\eta_{x_2 t} v_1 - \eta_{x_1 t} v_2\right] dx\, dt \\ &= \iint \left[\eta_{x_1 x_2}(v_2^2 - v_1^2) + (\eta_{x_2 x_2} - \eta_{x_1 x_1})(v_1 v_2)\right] dx\, dt.\end{aligned}$$

Thus, an incompressible velocity field, $v(x,t)$, defines a weak solution provided that the identity in (2.2) is satisfied (see [7]).

To establish the theorem, one needs to show that the regularized solutions, $v^\varepsilon(x,t)$, defined in (1.9) and (1.10) converge to a weak solution as $\varepsilon \to 0$. With the regularizations in (1.9) and (1.10), the family $v^\varepsilon(x,t)$ forms an approximate solution sequence for 2-D Euler with vortex sheet initial data in the precise sense of [6] (see pp. 316–318 and pp. 325–327 of [6]). Therefore, the functions v^ε and $\omega^\varepsilon = \operatorname{curl} v^\varepsilon$ satisfy the uniform bounds

$$(2.3)\qquad \max_{0\le t\le T}\left(\int_{R^2} |\omega^\varepsilon|\, dx + \int_{|x|\le R} |v^\varepsilon|^2\, dx\right) \le C_{R,T}$$

with constants $C_{R,T}$ uniform in ε for every $R, T > 0$. Also, Theorem 1.1 of [6] applies so that after passing to a subsequence there exists $v \in L^\infty\big([0,\infty), L^2_{\mathrm{loc}}(R^2)\big)$ with $\operatorname{curl} v = \omega \in L^\infty\big([0,\infty), M(R^2)\big)$ satisfying

$$\begin{aligned}&v^\varepsilon \to v \text{ weakly in } L^2(\Omega_{R,T}),\\ (2.4)\qquad &\omega^\varepsilon \to \omega \text{ weakly in } L^\infty\big([0,T], M(R^2)\big),\\ &v^\varepsilon \to v \text{ strongly in } L^p(\Omega_{R,T}),\ 1 \le p < 2,\end{aligned}$$

for any $T > 0$ and any space-time open set $\Omega_{R,T} = \{(x,t) \mid |x| < R,\ 0 < t < T\}$; furthermore, v satisfies the weak regularity condition in time required in (B) of (1.6) with $v(0) = v_0$ so that v assumes the required vortex sheet initial data. In addition, it follows from (1.10) that the vorticity ω^ε satisfies the scalar parabolic equation ([16])

$$\begin{aligned}(2.5)\qquad &\frac{\partial \omega^\varepsilon}{\partial t} + v^\varepsilon \cdot \nabla \omega^\varepsilon = \nu_\varepsilon \Delta \omega^\varepsilon,\\ &\omega^\varepsilon|_{t=0} = \omega_0^\varepsilon.\end{aligned}$$

Thus, if the initial vorticity has a distinguished sign, $\omega_0^\varepsilon \ge 0$, it follows from the maximum principle and (2.4) that

$$(2.6)\qquad \omega^\varepsilon(x,t) \ge 0, \quad \omega \ge 0,$$

so both $\omega^\varepsilon(x,t)$ and the limiting vorticity measure, ω, have a distinguished sign for any later time.

The smooth solutions in (1.10) are trivially weak solutions of either the Euler or Navier-Stokes equations. With these properties for v^ε (see pp. 325–327 of [6]), the weak convergence in (2.4), and the special structure for the weak form in (2.2), the limiting velocity, v, is a weak solution for 2-D Euler provided that the special nonlinear functions $v_2^2 - v_1^2$ and $v_1 v_2$ satisfy

$$\iint \eta_{x_1 x_2}(v_2^2 - v_1^2)\,dx\,dt = \lim_{\varepsilon\to 0}\iint \eta_{x_1 x_2}\big((v_2^\varepsilon)^2 - (v_1^\varepsilon)^2\big)\,dx\,dt \tag{2.7}$$

and

$$\iint (\eta_{x_2 x_2} - \eta_{x_1 x_1})(v_1 v_2)\,dx\,dt = \lim_{\varepsilon\to 0}\iint (\eta_{x_2 x_2} - \eta_{x_1 x_1}) v_1^\varepsilon v_2^\varepsilon\,dx\,dt. \tag{2.8}$$

The Navier-Stokes (Euler) equations are rotationally invariant, i.e., if $v^\varepsilon(x,t)$ solves these equations and suitable estimates, $\mathcal{O}(\theta)v^\varepsilon\big({}^t\mathcal{O}(\theta)x,t\big)$ satisfies these same equations and estimates for any rotation matrix $\mathcal{O}(\theta)$; through rotation by $45°$, the term $(v_2^\varepsilon)^2 - (v_1^\varepsilon)^2$ becomes $v_1^\varepsilon v_2^\varepsilon$ in the rotated reference frame (see [7]). Thus, in order to prove the theorem that v is a weak solution for 2-D Euler, it is sufficient to establish the weak convergence of $v_1^\varepsilon v_2^\varepsilon$ to $v_1 v_2$, i.e.,

$$\iint \phi v_1 v_2\,dx\,dt = \lim_{\varepsilon\to o}\iint \phi v_1^\varepsilon v_2^\varepsilon\,dx\,dt \tag{2.9}$$

for all test functions $\phi \in C_0^\infty\big(R^2 \times (0,\infty)\big)$. This reduction completes the first main part of the proof of the Theorem as described in (1.11) (1).

The circulation around any point x_0 enclosed inside a circle of radius R is given by

$$\int_{|x-x_0|\le R} \omega^\varepsilon(x,t)\,dx \tag{2.10}$$

Physical intuition suggests that if this circulation decays at a sufficiently rapid uniform rate as $R \to 0$ independent of ε then the limiting behavior of v^ε in L^2 can be controlled. Such a result of this type was developed in Theorem 3.1 of [6] where a condition for strong L^2-convergence of the velocity field was developed. The second ingredient from (1.11) in the proof of the theorem is the following:

Proposition (Uniform Decay of the Vorticity Maximal Function). *Assume that ω_0^ε, v_0^ε satisfy* (1.1)–(1.3) *and* (1.9) *and that ω_0^ε is nonnegative, $\omega_0^\varepsilon \geq 0$. Then the vorticity maximal functions defined through* (2.10) *have an a priori uniform decay rate and satisfy*

$$\max_{\substack{0\leq t\leq T\\ x_0\in R^2\\ 0<\varepsilon\leq\varepsilon_0}} \int_{|x-x_0|\leq R} \omega^\varepsilon(x,t)\,dx \leq C_{T,R_0}|\log(2R)|^{-1/2}, \qquad \text{for } R<\frac{1}{2}, \tag{2.11}$$

where C_{T,R_0} is a fixed constant depending on T, R_0 from (1.3) *and the uniform bounds in* (1.9) *and* (2.3). *The same estimate in* (2.11) *also applies to the limiting vorticity measure, $\omega = \operatorname{curl} v$ from* (2.4), *in fact,*

$$\max_{\substack{0\leq t\leq T\\ x_0\in R^2}} \int_{|x-x_0|\leq R} d\omega(x,t) \leq C_{T,R_0}|\log(4R)|^{-1/2}, \qquad \text{for } R<\tfrac{1}{4}. \tag{2.12}$$

The uniform a priori estimate in (2.11) is sharp for initial measures with a distinguished sign, $\omega_0^\varepsilon \geq 0$, and false for general initial vorticity distributions without a distinguished sign (see Section 3). The proof of this estimate in (2.11) is given in detail in Section 3 below; this proposition is the second main fact needed in the proof of the theorem.

In order to pass to the limit as $\varepsilon \to 0$ and obtain a weak solution of 2-D Euler, it follows from (2.9) that one needs to understand the expression,

$$\int_{R^2} \varphi v_1 v_2\,dx \qquad \text{for any } \varphi \in C_0^\infty(R^2).$$

Utilizing the fact from (1.8) that the velocity is determined from the vorticity by convolution with the mildly singular integral in (1.8) results in the identity

$$\int_{R^2} \varphi v_1 v_2\,dx = \iint_{R^2\times R^2} H_\varphi \omega(x)\omega(y)\,dx\,dy, \tag{2.13}$$

where $H_\varphi(x,y)$ is the function (distribution) on $R^2\times R^2$ given by the formula

$$H_\varphi(x,y) = -\frac{1}{4\pi}\frac{\partial^2}{\partial x_1 \partial x_2}\int_{R^2} \log|x-z|\log|y-z|\varphi(z)\,dz. \tag{2.14}$$

In general, when kernels that are homogeneous of degree -1 in R^2 are iterated to build a function such as $H_\varphi(x,y)$, logarithmic singularities occur along the diagonal, $x=y$. Delort's *key new observation* in [10] (see Proposition 1.2.3 in [10]) is the important fact that with the *special nonlinearities* v_1v_2, such iterated singular integrals result in a *function*, H_φ, that remains *bounded along the diagonal*. The detailed statement of his result is the following:

Delort's Lemma. *The function $H_\varphi(x,y)$ defined in* (2.14) *is a bounded function on $R^2 \times R^2$, continuous on the complement of the diagonal, and tends to zero at infinity. In fact,*

$$H_\varphi(x,y) = \frac{1}{2}\big(\varphi(x)+\varphi(y)\big)h(x-y)+r(x,y) \tag{2.15}$$

with $h(\omega) = \omega_1 \cdot \omega_2/4\pi|\omega|^2$ and $r(x,y)$, a bounded continuous function tending to zero at infinity.

Delort proves this lemma by direct calculation in a few paragraphs following Proposition 1.2.3 in [10] and I refer the interested reader to that work for the details. Next, with the three main facts in (2.9), (2.11)–(2.12), and (2.13)–(2.15), I pass to the limit in (2.9) and verify that v is a weak solution of 2-D Euler. Through a standard density argument, it is no loss of generality to verify (2.9) only for the special product test functions

$$\phi(x,t) = \psi(t)\varphi(x), \quad \text{with } \psi(t) \in C_0^\infty(R^+),\ \varphi(x) \in C_0^\infty(R^2). \tag{2.16}$$

Let $\rho(|x|) \in C_0^\infty(R^2)$ be a fixed positive cut-off function with $\rho \equiv 1$ for $|x| < 1$ and $\rho \equiv 0$ for $|x| > 2$. With (2.13)

$$\begin{aligned} &\iint \phi(x,t)v_1^\varepsilon v_2^\varepsilon \, dx\, dt \\ &= \int_0^\infty \int_{R^2\times R^2} \int \psi(t)\left(1-\rho\left(\frac{|x-y|}{\delta}\right)\right) H_\varphi(x,y)\omega^\varepsilon(x,t)\omega^\varepsilon(y,t)\, dx\, dy\, dt \\ &\quad + \int_0^\infty \int_{R^2\times R^2} \int \psi(t)\rho\left(\frac{|x-y|}{\delta}\right) H_\varphi(x,y)\omega^\varepsilon(x,t)\omega^\varepsilon(y,t)\, dx\, dy\, dt. \end{aligned} \tag{2.17}$$

For any fixed δ with $1 > \delta > 0$, from Delort's lemma

$$\psi(t)\left(1-\rho\left(\frac{|x-y|}{\delta}\right)\right) H_\varphi(x,y) \tag{2.18}$$

is a continuous function tending to zero at infinity, and from (2.4) it follows directly that

$$\begin{aligned} &\omega^\varepsilon(x,t)\otimes\omega^\varepsilon(y,t) \to d\omega(x,t)\otimes d\omega(y,t) \text{ weakly in the measures,} \\ &L^\infty\big([0,T], M(R^2\times R^2)\big) \text{ for any } T>0. \end{aligned} \tag{2.19}$$

The facts in (2.18), (2.19) combine with the definition of weak convergence to yield

$$(2.20)\quad \lim_{\varepsilon\to 0}\int_0^\infty \int_{R^2\times R^2}\int \psi(t)\left(1-\rho\left(\frac{|x-y|}{\delta}\right)\right) H_\varphi(x,y)\,\omega^\varepsilon(x,t)\,\omega^\varepsilon(y,t)\,dx\,dy\,dt$$
$$= \int_0^\infty dt \int_{R^2\times R^2} \psi(t)\left(1-\rho\left(\frac{|x-y|}{\delta}\right)\right) H_\varphi(x,y)\,d\omega(x,t)\otimes d\omega(y,t).$$

Next, I utilize the uniform decay rate for the vorticity maximal function in (2.11) together with Delort's lemma to get uniform control on the second term on the right hand side of (2.17) resulting in the estimates,

$$(2.21)\quad \left|\int_0^\infty \int_{R^2\times R^2}\int \psi(t)\rho\left(\frac{|x-y|}{\delta}\right) H_\varphi(x,y)\,\omega^\varepsilon(x,t)\,\omega^\varepsilon(y,t)\,dx\,dy\,dt\right|$$
$$\le C_{\varphi,\psi}\int_0^T \int_{R^2\times R^2} \rho\left(\frac{|x-y|}{\delta}\right)\omega^\varepsilon(x,t)\,\omega^\varepsilon(y,t)\,dx\,dy\,dt$$
$$\le C_{\varphi,\psi}C_{R_0,T}\left(\log\left(\frac{1}{2\delta}\right)\right)^{-1/2}\int_0^T\int_{R^2}\omega^\varepsilon(y,t)\,dy\,dt$$
$$\le C\left(\log\left(\frac{1}{2\delta}\right)\right)^{-1/2}.$$

Delort's lemma and the nonnegative character of ω^ε have been used in the first inequality in (2.21) while the uniform decay in (2.11) for the vorticity maximal function has been applied in the second inequality in (2.21); finally, the uniform control on the mass of the measures, ω^ε, guaranteed by (2.3) has been applied in the final inequality in (2.20). With (2.12), (2.3), and (2.4), the same uniform estimate in (2.21) applies with ω^ε replaced by the limiting vorticity measure, ω. Thus, (2.20) and (2.21) guarantee that

$$(2.22)\quad \left|\iint \phi(x,t)v_1v_2\,dx\,dt - \lim_{\varepsilon\to 0}\iint \phi v_1^\varepsilon v_2^\varepsilon\,dx\,dt\right| \le C_\phi\left(\log\left(\frac{1}{4\delta}\right)\right)^{-1/2}.$$

Since δ in (2.22) can be made arbitrarily small, the requirement in (2.9) is satisfied and v is a weak solution of the 2-D Euler equations with the required vortex sheet initial data, $v_0(x)$. This completes the proof of the theorem. It is worth mentioning here that the distinguished sign for the vorticity is only utilized in two places in the proof: to derive the proposition on uniform decay of the vorticity maximal function and in the second step in the inequality in (2.21).

3. A priori uniform decay for the vorticity maximal function with nonnegative vorticity. Here I prove the proposition involving the estimates in (2.11), (2.12) which were utilized in the proof of the theorem. One subtle feature of fluid flow in R^2 with nonnegative vorticity is the fact that the velocity field has local finite kinetic energy but $v \notin L^2(R^2)$. In fact for vorticity $\omega(x) \in C_0^\infty(R^2)$, the Biot-Savart law in (1.8) implies that for x with $|x| \gg 1$,

$$v(x) \cong \frac{1}{2\pi}|x|^{-2}\begin{pmatrix} -x_2 \\ x_1 \end{pmatrix}\int_{R^2} \omega(y)\,dy + o(|x|^{-1}) \tag{3.1}$$

so that $v(x) \notin L^2(R^2)$ for nonnegative vorticity, $\omega \geq 0$. The natural replacement for the kinetic energy in this context is the "pseudo energy" defined through the stream function ψ from (1.7) by

$$-\int \psi\omega = -\frac{1}{2\pi}\int_{R^2\times R^2}\int \log|x-y|\,\omega(x)\,\omega(y)\,dx\,dy \equiv H(\omega). \tag{3.2}$$

For smooth velocity fields $v(x,t) \in L^2(R^2)$, it follows from integration by parts that $\int v\cdot v = -\int \psi\omega$ so that the pseudo energy generalizes the kinetic energy for velocity fields with $v \notin L^2$ (such as always occurs for $\omega \geq 0$). For smooth solutions of the 2-D Euler equations with initial vorticity, ω_0, of compact support, the "pseudo energy" is a conserved quantity in the dynamics, i.e.,

$$H\big(\omega(t)\big) = H\big(\omega(0)\big), \qquad \text{for all } t. \tag{3.3}$$

It is an exercise for the reader to verify this by utilizing the Euler equations in (1.10) and (2.5) with $\nu_\varepsilon = 0$. It is worth mentioning here that computational vortex blob methods exactly conserve a discrete version of the function $H(\omega)$ in (3.2) (see [6] p. 332 eqn. (2.31)) and this discrete function serves as the Hamiltonian for smoothed core discrete vortex dynamics; since vortex blob algorithms converge for smooth solutions of the Euler equations (see [4]), one can pass to the continuous limit and verify the conserved quantity in (3.3). Hopefully, these remarks provide some intuition for the reader regarding the pseudo energy functional, $H(\omega)$, in ideal fluid flow.

One well known conserved quantity for smooth solutions of the Navier-Stokes equations and easily derived from (2.5) is the total vorticity; thus

$$N\big(\omega(t)\big) \equiv \int_{R^2} \omega(x,t)\,dx = \int_{R^2} \omega(x,0)\,dx \qquad \text{for all } t \geq 0. \tag{3.4}$$

Another useful physical conserved quantity is the second moment of vorticity,

$$L(\omega) = \int_{R^2} |x|^2\omega(x)\,dx. \tag{3.5}$$

With the vorticity form for the Navier-Stokes equations in (2.5), it is a simple matter to integrate by parts for solutions with smooth rapidly decreasing vorticity and obtain the formula

$$L(\omega(t)) = L(\omega(0)) + 2\nu_\varepsilon t N(\omega(0)). \tag{3.6}$$

Thus, the second moment of vorticity is exactly conserved by the 2-D Euler equations and grows slowly with time for solutions of the Navier-Stokes equations through viscous diffusion.

Combining (3.3) and (3.6), it follows that smooth solutions of the Euler equation conserve the two quantities, $H(\omega)$ and $L(\omega)$. This motivates the following lemma which will immediately imply the estimates in (2.11), (2.12) of the proposition in the special case that v^ε in (1.10) satisfies the 2-D Euler equations:

Lemma. *Assume that ω is a smooth* nonnegative *vorticity function, i.e., $\omega \geq 0$, and that*

$$\int_{R^2} \omega\, dx \leq \bar{M}, \quad |H(\omega)| \leq \bar{H}, \quad |L(\omega)| \leq \bar{L}. \tag{3.7}$$

Then the vorticity maximal function satisfies the decay rate

$$\max_{x_0 \in R^2} \int_{|x-x_0| \leq R} \omega(x)\, dx \leq C_{\bar{M},\bar{H},\bar{L}} (|\log(2R)|)^{-1/2}, \qquad \text{for } R < \frac{1}{2}, \tag{3.8}$$

with the a priori constant, $C_{\bar{M},\bar{H},\bar{L}}$ given by $C_{\bar{M},\bar{H},\bar{L}} = (\bar{H} + 2\bar{L}\bar{M})^{1/2}$.

Proof. Set $\log^+(s) = \log(s)$ for $s > 1$ and $\log^+(s) = 0$ for $s \leq 1$ and define $\log^-(s) = \log(s) - \log^+(s)$; then simple estimates yield

$$\begin{aligned} &\int_{R^2 \times R^2} \int \left|\log^-(|x-y|)\right| \omega(x)\omega(y)\, dx\, dy \\ &\leq |H(\omega)| + \int_{R^2 \times R^2} \int \log^+(|x-y|)\omega(x)\omega(y)\, dx\, dy \\ &\leq |H(\omega)| + 2\int_{R^2 \times R^2} \int (|x|^2 + |y|^2)\omega(x)\omega(y)\, dx\, dy \\ &\leq \bar{H} + 2\bar{L}\bar{M}. \end{aligned} \tag{3.9}$$

Also, for $R < \frac{1}{2}$, one obtains

$$\begin{aligned} &\int_{R^2 \times R^2} \int \left|\log^-(|x-y|)\right| \omega(x)\omega(y)\, dx\, dy \\ &\geq \int_{|x-x_0| \leq R} \int_{|y-x_0| \leq R} \left|\log^-(|x-y|)\right| \omega(x)\omega(y)\, dx\, dy \\ &\geq |\log(2R)| \left(\int_{|x-x_0| \leq R} \omega(x)\, dx \right)^2. \end{aligned} \tag{3.10}$$

Combining (3.9) and (3.10) yields the required estimate in (3.8) with the explicit constant $C_{\bar{M},\bar{H},\bar{L}} = (\bar{H} + 2\bar{L}\bar{M})^{1/2}$.

Next, I show how the lemma above directly yields the proof of the proposition for the special case when the approximating solutions, v^ε, from (1.10) are smooth solutions of the 2-D Euler equations so that $\nu_\varepsilon \equiv 0$ in (1.10). (This is the only case treated by Delort in [10]). For the 2-D Euler equations, the quantities $\int_{R^2} \omega(t)\,dx$, $H\big(\omega(t)\big)$, $L\big(\omega(t)\big)$ are all exactly conserved for all times $t \geq 0$ and have the same value as defined at time $t = 0$. Thus, to deduce the estimate in (2.11) from (3.8) in the lemma, it is sufficient to obtain uniform bounds independent of ε in the initial data for the functionals $\int_{R^2} \omega_0^\varepsilon(x)\,dx$, $L(\omega_0^\varepsilon)$, $|H(\omega_0^\varepsilon)|$. With the regularization described in (1.9) and (3.6), the ε-independent bounds for $\int_{R^2} \omega_0^\varepsilon(x)\,dx$ and $L(\omega_0^\varepsilon)$ are obvious; in particular, the approximations, ω_0^ε, satisfy

$$\int_{R^2} \omega_0^\varepsilon(x)\,dx = \int_{R^2} d\omega_0 \equiv \Gamma_0. \tag{3.11}$$

The uniform bound for $|H(\omega_0^\varepsilon)|$ is slightly more tedious and involves similar considerations as were utilized earlier in [6] (see pp. 317 and 325–327 of [6]) to handle the fact in (3.1) that $v^\varepsilon \notin L^2(R^2)$ when ω_0^ε is nonnegative. I sketch the argument for completeness. As on page 317 of [6], choose a fixed radial function $\bar{\omega}_0(r) \in C_0^\infty(R^2)$ with supp $\bar{\omega}_0(r) \subseteq \{|x| < R_0\}$ and

$$\Gamma_0 = 2\pi \int_0^\infty r\bar{\omega}_0(r)\,dr. \tag{3.12}$$

The induced velocity,

$$\bar{v}_0(x) = |x|^{-2} \begin{pmatrix} -x_2 \\ x_1 \end{pmatrix} \int_0^{|x|} s\bar{\omega}(s)\,ds, \tag{3.13}$$

is the corresponding radial eddy defining an exact solution of the 2-D Euler equations (see [16]) with $\bar{\psi}(x)$ the corresponding stream function. It is an elementary exercise to show that $\bar{\psi}(x)$ satisfies the bound

$$|\bar{\psi}(x)| \leq C\big(1 + \log^+(|x|)\big). \tag{3.14}$$

Consider the velocity field $\tilde{v}_0^\varepsilon = v_0^\varepsilon - \bar{v}$; with the conditions in (3.11) and (3.12), a standard far-field calculation refining (3.1) yields the uniform estimate,

$$\int_{|x|>2R_0} |\tilde{v}_0^\varepsilon|^2 \leq C\left(\int_{|x|\leq R_0} |\tilde{\omega}_0^\varepsilon|\,dx\right)^2 \leq 4C\Gamma_0^2, \tag{3.15}$$

so that (1.9), (3.13), and (3.15) guarantee the uniform L^2-velocity bound

$$\max_{\varepsilon \le 1} \int_{R^2} |\tilde{v}_0^\varepsilon|^2 \, dx \le C. \tag{3.16}$$

To establish the required uniform bound on $H(\omega_0^\varepsilon)$, decompose $H(\omega_0^\varepsilon)$ as follows,

$$\begin{aligned} H(\omega_0^\varepsilon) &= H(\tilde{\omega}_0^\varepsilon) + H(\bar{\omega}_0) - 2\int_{R^2} (\tilde{\omega}_0^\varepsilon \bar{\psi}) \, dx \\ &= \int_{R^2} |\tilde{v}_0^\varepsilon|^2 \, dx + H(\bar{\omega}_0) - 2\int_{R^2} (\tilde{\omega}_0^\varepsilon \bar{\psi}) \, dx. \end{aligned} \tag{3.17}$$

In the second identity in (3.17), I utilized the fact mentioned below (3.2) that for $\tilde{v} \in L^2$ with vorticity $\tilde{\omega}$ rapidly decreasing,

$$\int_{R^2} |\tilde{v}|^2 = -\int_{R^2} \tilde{\psi}\tilde{\omega}. \tag{3.18}$$

With (3.16) the first two terms on the right hand side of (3.17) are uniformly bounded while with (3.14) the remaining term satisfies

$$\left| \int_{R^2} \tilde{\omega}_0^\varepsilon \bar{\psi} \, dx \right| \le C\Gamma_0. \tag{3.19}$$

Thus, $|H(\omega_0^\varepsilon)| \le \bar{H}$ as required in the lemma. It is an elementary exercise for the reader to verify that with the weak convergence as measures of ω^ε to ω in (2.4), the measure ω automatically inherits the slightly weaker bound in (2.12) from the uniform bound on the vorticity maximal functions for ω^ε in (2.11). This completes the proof of the proposition in the case that $v^\varepsilon(x,t)$ is a sequence of smooth solutions satisfying the 2-D Euler equations.

To extend the above argument applying the lemma to an approximating sequence, v^ε, involving smooth solutions of the Navier-Stokes equations, it is necessary to establish ε-independent bounds on the three functionals $\int_{R^2} \omega^\varepsilon(x,t) \, dx$, $L(\omega^\varepsilon(t))$, $H(\omega^\varepsilon(t))$ on the fixed time interval, $0 \le t \le T$ for an arbitrary $T > 0$. The properties in (3.4), (3.6), and (3.11) combine with the fact that $L(\omega_0^\varepsilon)$ is uniformly bounded to guarantee the required ε-independent bounds for $\int_{R^2} \omega^\varepsilon(x,t) \, dx$ and $L(\omega^\varepsilon(t))$ on any finite time interval. To establish the ε-independent bound for $H(\omega^\varepsilon(t))$, I utilize a similar device as was used earlier on p. 326 of [6]. Let $\bar{\omega}^\varepsilon(|x|,t)$ be the radial solution of the diffusion equation,

$$\begin{aligned} &\frac{\partial \bar{\omega}^\varepsilon}{\partial t} = \nu_\varepsilon \Delta \bar{\omega}^\varepsilon, \\ &\bar{\omega}^\varepsilon|_{t=0} = \bar{\omega}_0(|x|). \end{aligned} \tag{3.20}$$

with the initial data $\bar{\omega}_0(|x|)$ defined above (3.12). It follows (see [16]) that

$$\bar{v}^\varepsilon(x,t) = |x|^{-2}\begin{pmatrix} -x_2 \\ x_1 \end{pmatrix}\int_0^{|x|} s\bar{\omega}^\varepsilon(s,t)\,ds \tag{3.21}$$

defines an exact solution of the Navier-Stokes equations with

$$\begin{aligned} \frac{\partial \bar{v}^\varepsilon}{\partial t} &= \nu_\varepsilon \Delta \bar{v}^\varepsilon, \\ \bar{v}^\varepsilon \cdot \nabla \bar{v}^\varepsilon &= -\nabla \bar{p}^\varepsilon \end{aligned} \tag{3.22}$$

Furthermore, the stream functions $\bar{\psi}^\varepsilon$ and velocity fields $\bar{v}^\varepsilon$ satisfy the uniform bound,

$$\begin{aligned} &\max_{\substack{0\le t\le T\\ 0<\varepsilon\le 1}} |\bar{\psi}^\varepsilon(|x|,t)| \le C(1+\log^+|x|), \\ &\max_{\substack{0\le t\le T\\ 0<\varepsilon\le 1}} |\nabla \bar{v}^\varepsilon(t)|_{L^\infty} \le C. \end{aligned} \tag{3.23}$$

Consider $\tilde{v}^\varepsilon = v^\varepsilon(x,t) - \bar{v}^\varepsilon(x,t)$; since both $\omega^\varepsilon(x,t)$ and $\bar{\omega}^\varepsilon(x,t)$ are rapidly decreasing and the total vorticity is conserved in Navier-Stokes solutions (see (3.4)), it follows by similar far field arguments as presented earlier that $\tilde{v}^\varepsilon(x,t) \in L^2(R^2)$ for any fixed ε. With (3.22) and (1.10) the function $\tilde{v}^\varepsilon$ satisfies the equations

$$\begin{aligned} \frac{D\tilde{v}^\varepsilon}{Dt} + (\tilde{v}^\varepsilon \cdot \nabla)\bar{v}^\varepsilon(x,t) &= -\nabla \tilde{p}^\varepsilon + \nu_\varepsilon \Delta \tilde{v}^\varepsilon, \\ \operatorname{div} \tilde{v}^\varepsilon &= 0. \end{aligned} \tag{3.24}$$

Standard energy estimates obtained by multiplying (3.24) by $\tilde{v}^\varepsilon$ and using (3.23) together with Gronwall's inequality establish that

$$\max_{0\le t\le T}\int_{R^2} |\tilde{v}^\varepsilon|^2(x,t)\,dx \le C(T)\int_{R^2} |\tilde{v}_0^\varepsilon|^2\,dx \le \tilde{C}(T). \tag{3.25}$$

Thus, the inequalities in (3.25) together with (3.16) guarantee an ε-independent bound for the energy of the velocity fields, $\tilde{v}^\varepsilon(x,t)$. As in (3.17), one has

$$H\big(\omega^\varepsilon(t)\big) = \int_{R^2} |\tilde{v}^\varepsilon|^2(t)\,dx + H\big(\bar{\omega}^\varepsilon(t)\big) - 2\int_{R^2} \omega^\varepsilon(t)\bar{\psi}^\varepsilon(t)\,dx. \tag{3.26}$$

With the estimate in (3.23) and (3.6), it follows that

$$\left|\int_{R^2} \omega^\varepsilon(x,t)\bar{\psi}^\varepsilon(x,t)\,dx\right| \le C\int_{R^2}(1+|x|^2)\omega^\varepsilon(x,t)\,dx \le C\big(\Gamma_0 + L_0 + \nu_\varepsilon t N(\omega_0)\big). \tag{3.27}$$

Clearly, (3.25)–(3.27) all combine to yield the ε-independent bound,

$$\max_{\substack{0\le t\le T\\ 0<\varepsilon\le 1}} \big|H\big(\omega^\varepsilon(t)\big)\big| \le C(T)$$

as required to apply the lemma. This fact completes the proof of the proposition from Section 2.

Examples illustrating the sharp features of the bounds. Despite the elementary proofs for the estimates in (2.11) and (3.8), here I present some simple examples which illustrate that these bounds in (2.11) are sharp. As preliminary information, I recall the following fact (see [16]):

If $\omega(|x|)$ is an arbitrary smooth radial function, then

$$v(x) = \begin{pmatrix} -x_2 \\ x_1 \end{pmatrix} |x|^{-2} \int_0^{|x|} s\omega(s)\,ds \tag{3.28}$$

defines an exact solution of the 2-D Euler equations.

▷ EXAMPLE #1. Pick a radial vorticity function, $\omega(|x|) \ge 0$ with $\omega(|x|) \in C_0^\infty(R^2)$. Define the rescaled velocity fields

$$v^\varepsilon(x) = \left(\log\left(\frac{1}{\varepsilon}\right)\right)^{-1/2} \varepsilon^{-1} v\left(\frac{x}{\varepsilon}\right) \tag{3.29}$$

with v given in (3.28). Then $v^\varepsilon(x)$ *is a sequence of exact solutions with nonnegative vorticity with uniform bounds on both the* L^1*-norm of the vorticity* and the *local kinetic energy* as required in (2.3). (See the detailed calculations in [5].) An explicit calculation establishes that

$$\max_{0<\varepsilon\le 1} \int_{|x|\le R} \omega^\varepsilon(x)\,dx \ge C|\log(R)|^{-1/2} \tag{3.30}$$

with $C > 0$. Thus, *there are exact solution sequences with nonnegative vorticity so that the bounds* in (2.11) *are sharp.* If one recalls the key estimate in (2.20), it becomes natural to ask whether there are uniform decay rates for the quantity

$$\max_{x_0\in R^2} \int_{|x-x_0|\le R} |\omega^\varepsilon(x)|\,dx$$

when the velocity and vorticity satisfy the uniform bounds in (2.3) but the vorticity, $\omega^\varepsilon(x)$, does not have a distinguished sign. This possibility leads to:

▷ EXAMPLE #2. Pick a radial function $\omega(r) \in C_0^\infty(R^2)$ so that

$$\int_0^\infty s\omega(s)\,ds = 0.$$

Define $v^\varepsilon(x)$ in this case by

$$v^\varepsilon(x) = \varepsilon^{-1} v\left(\frac{x}{\varepsilon}\right) \tag{3.31}$$

with $v(x)$ given in (3.28). Then $v^\varepsilon(x)$ is a sequence of exact solutions of the 2-D Euler equations satisfying the uniform vorticity and kinetic energy bounds in (2.3) (see [6]). An elementary calculation establishes that

$$\max_{0<\varepsilon\le 1} \int_{|x|\le R} |\omega^\varepsilon(x)|\,dx \ge C > 0. \tag{3.32}$$

Thus, $v^\varepsilon(x)$ is a sequence of exact solutions where the vorticity does not have a distinguished sign and there is no a priori uniform decay rate for the vorticity maximal function.

These examples illustrate that much more complex phenomena can occur when the vorticity changes sign. Such behavior is confirmed by careful numerical simulations ([3], [4]).

4. Concluding remarks. This paper studies the convergence to weak solutions for the 2-D Euler equations with vortex sheet initial data provided that the vorticity has a distinguished sign. Theorem 3.1 from [6] guarantees that the velocity fields, v^ε, converge *strongly in* L^2 to a *weak solution* of 2-D Euler when the *vorticity* is *nonnegative* provided that the vorticity maximal functions decay at the uniform rate,

$$\max_{\substack{x_0\in R^2\\ 0\le t\le T\\ 0<\varepsilon\le 1}} \int_{|x-x_0|\le R} \omega^\varepsilon(x,t)\,dx \le C\big(|\log(R)|\big)^{-(1+\beta)}, \tag{4.1}$$

for some positive constant $\beta > 0$. The proposition proved in Section 3 with the estimate in (2.11) indicates that there is a very slight discrepancy between the general behavior of an approximate solution sequence with nonnegative vorticity and the situation with strong convergence to a weak solution—the general a priori estimate in (2.11) has the logarithmic decay exponent $\frac{1}{2}$ rather than $1+\beta$ with $\beta > 0$ from (4.1). Nevertheless, as the examples in (3.29) indicate (see [5]), concentration phenomena without strong convergence can occur when the

vorticity has a distinguished sign and the proof presented in Section 2 yields a weak solution for 2-D Euler despite such concentration effects. It is worth emphasizing once again that there is strong evidence through explicit examples for an analogous but simpler model problem involving the 1-D Vlasov-Poisson equations ([12], [13]) that the weak solutions in the theorem probably are not unique and depend on the regularization even though the vorticity has a distinguished sign. It would be very interesting to make significant progress on this possible nonuniqueness of weak solutions for the 2-D Euler equations with vortex sheet initial data.

As mentioned earlier in the introduction, both numerical calculations and various theoretical examples ([3], [4], [5], [6], [7], [17]) indicate much more complex behavior in two-dimensional vortex sheet evolution without a distinguished sign. The main theoretical results involve "concentration-cancellation" yielding weak solutions in the steady case ([5]) together with various refinements ([5], [17], [18], [19]). Measure-valued solutions with a special "concentration" structure were constructed in [6] and it was conjectured already in [4] that perhaps such measure-valued solutions are really needed in some circumstances for vortex sheets when the vorticity changes sign. The 1-D Vlasov-Poisson equations for a two-component plasma have an analogous structure to the behavior of the 2-D Euler equations when the vorticity changes sign (see [13], [14]). In [14] simple explicit examples with dynamically evolving concentrations are constructed where *perturbed weak solutions dynamically converge to a measure-valued solution for the 1-D Vlasov Poisson equations which is not a weak solution.* These examples provide further supporting evidence regarding the possibility of a significant role for measure-valued solutions in limiting procedures for 2-D vortex sheets with vorticity that changes sign. Obviously, it is a major open problem to understand the behavior of weak solutions and measure-valued solutions for the 2-D Euler equation and general 2-D vortex sheets without a distinguished sign.

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